acyclovir

Warnings	Classification	Alternate Names
irritant (reports of vesicant effects)	antiviral agent	ZOVIRAX acyclovir sodium

Indications

- treatment of mucocutaneous and disseminated infections caused by herpes simplex virus (HSV) types 1 and 2 and varicella zoster (VZ)
- prophylaxis of HSV in immunocompromised patients

Reconstitution and Stability

- stable at room temperature

Compatibility

- compatible with D5W, NS, dextrose-saline solutions and lactated ringers

Preparation and Administration

Administration Route	Approved	Preparation and Administration Instructions	Required Monitoring
Subcutaneous	NO		
Intramuscular	NO		
IV direct	NO		
IV intermittent	YES	use large vein in forearm; if possible avoid hand, wrist or antecubital fossa dilute doses: ○ up to 700 mg in 100 mL ○ greater than 700 mg in 250 mL infuse over 1 hour	Basic Monitoring
Continuous IV infusion	NO		

© 2023 BC Women's Hospital + Health Centre Providence Health Care Vancouver Coastal Health All rights reserved
Refer to intranet for current version

acyclovir

Dosage

- Use total body weight (TBW) for dosing.
  If TBW is more than 125% IBW (ideal body weight), use [adjusted body weight = IBW + 0.4 (TBW - IBW)] to calculate the dose.

- HSV infection: 5 mg/kg IV Q8H for 7 days

- Herpes encephalitis, herpes ophthalmitis, VZ: 10 to 15 mg/kg IV Q8H for 7 to 14 days

- Bone Marrow Transplant HSV prophylaxis: 5 mg/kg IV Q12H

- Dose adjustment in renal impairment as follows:

Creatinine Clearance	Dose
30 to 50 mL/minute	5 to 10 mg/kg every 12 hours
Less than 30 mL/minute	5 to 10 mg/kg every 24 hours
Hemodialysis (HD)	2.5 to 5 mg/kg every 24 hours after dialysis on HD days
Continuous ambulatory peritoneal dialysis	2.5 to 5 mg/kg every 24 hours

Potential Hazards of Parenteral Administration

- phlebitis at the injection site
- rapid intravenous injection may result in precipitation of acyclovir crystals in the renal tubules
- Extravasation risk: irritant (reports of vesicant effects); at concentration greater than 7 mg/mL
  - **refer to** : Extravasation Management (Non-Antineoplastic Vesicant/ Irritant Medications) – Adults

Important Implications

Contraindications/Cautions

- Contraindication: patients who have hypersensitivity to acyclovir or valacyclovir
- *Caution*:
  - an increase in serum creatinine and decreased creatinine clearance may occur in patients who:
    - are poorly hydrated, ensure adequate hydration (e.g. NS 250 mL to 500 mL) prior to each dose, based on volume status
    - are receiving concomitant nephrotoxic drugs
    - have pre-existing renal damage
- Pregnancy/Lactation: refer to Lexicomp or Micromedex

Side effects

- rash, urticaria, diaphoresis, hematuria, encephalopathic changes and nausea may occur
- increase in serum creatinine

© 2023 BC Women's Hospital + Health Centre Providence Health Care Vancouver Coastal Health
All rights reserved
Refer to intranet for current version